

Capital account liberalisation in India: An analysis of onshore-offshore arbitrage

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Abstract

In this paper we measure the capital account openness in India, a country with an elaborate and comprehensive system of capital controls covering all kinds of international financial transactions. We cover a span of 20 years and analyse the daily return differentials in the non-deliverable currency forward (NDF) market. We calculate the deviations from the covered interest parity (CIP) and estimate structural breaks in these deviations. We then connect the changes over time in the CIP deviations to the patterns of changes in de-jure capital control actions implemented by the Indian authorities. We also estimate no-arbitrage bands using a Self-Exciting Threshold Autoregressive (SETAR) model. We find that India has become more financially integrated with the global markets in recent years, as demonstrated by smaller CIP deviations and narrowing arbitrage bands.

JEL classification: G15, F30, F31, F32

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1 Introduction

The debate around the merits and demerits of allowing cross-border flows of financial capital is far from settled. Those in favour of capital account liberalisation claim that free movement of capital across national borders allows more efficient allocation of resources which in turn improves productivity and economic growth in all countries concerned. It also facilitates diversification of risk. Financial integration with the global economy brings many opportunities such as providing access to foreign savings which can be used to finance domestic investment (Schindler, 2009; Prasad, India paper). Critics on the other hand highlight the risks associated with large and volatile cross-border capital flows which may lead to economic distortions and policy challenges. Sudden reversal of capital flows could trigger financial instability in the recipient countries and surge of foreign capital could cause asset bubbles.

In the last three decades, the magnitude of cross-border capital flows has gone up by leaps and bounds as most emerging market economies (EMEs) actively took steps to systematically dismantle their capital controls and open up their economies to foreign investments. Yet from time to time these EMEs keep experimenting with capital control policies to satisfy a myriad objectives such as, limiting exchange rate appreciation or preventing a sharp depreciation, increasing monetary policy independence, reducing inflation, dealing with volatility in financial markets and so on Forbes (2015).

One important and interesting question in this context is whether and to what extent changes in capital controls lead to changes in the financial integration of an economy with the rest of the world. Before getting into the debate as to whether capital account liberalisation is hurting or helping an economy, it is important to first understand how open or closed a country's capital account truly is. In other words, what has been the *de-facto* outcome of the *de-jure* capital control actions taken by countries over time in response to global financial flows? Measuring the extent of integration with the rest of the world may help evaluate the effectiveness of the policies implemented.

In this paper we attempt to measure the changes over time in the capital account openness of a large and important emerging economy, India which continues to be prime exponent of a comprehensive capital controls system. India is perhaps the only country in the world, other than China, that till date has in place a complex and elaborate framework of capital controls. Despite adoption of liberalisation reforms in the 1990s, the overall capital controls system was never dismantled in India.

The Indian authorities keep changing their capital control policies on a fairly regular basis. Since the 1990s, the authorities have been gradually relaxing the legal restrictions that govern cross-border capital flows, but from time to time new restrictions are also imposed on the foreign investors. The frequent changes in capital controls make India a good case-study to analyse capital account liberalisation, and the manner in which it has progressed over the years. However, this also implies that measuring the extent of India's financial integration with the global economy is far from a straightforward exercise. Unlike many other EMEs, one cannot characterise India's capital account as either open or closed at any given point of time. Rather the extent of India's financial integration has to be deciphered on a continuous basis from available data.

Understanding how open a country's capital account is also has important policy implications in the context of Impossible Trilemma, a fundamental tenet of open economy macroeconomics. The Trilemma faced by all open economies is that no country can pursue independent monetary policy, stable exchange rates and capital account liberalisation at the same time. The relationship between domestic monetary policy and currency policy therefore critically hinges on the extent of the country's interaction with the international financial markets.

There are two ways to measure capital account openness of a country: de-jure measures and de-facto measures. De-jure measures are constructed from the legal restrictions by collecting data on changes in regulations.

Some of the relatively older cross-country de-jure measures are the Chinn-Ito index (Chinn and Ito (2008)) or the index constructed by Schindler (2009). These measures use the detailed capital controls published in the summary classifications table by the International Monetary Fund in the Annual Report on Exchange Arrangements and Exchange Restrictions (AREAER).

There exists a more recent literature that attempts to analyse the impact of capital control actions (CCAs) on various macroeconomic and financial variables. Most of these studies also construct some de-jure measure or index of capital account openness. For instance, Forbes, Fratzscher and Straub (2015) analyse the motivations for imposing capital controls by constructing a dataset that tracks increases and decreases in controls on capital inflows, controls on capital outflows, and macro-prudential measures at a weekly frequency for 60 countries from 2009 through 2011. Pasricha et al. (2018) investigate the effectiveness of CCAs and examine their spillover effects using a high frequency dataset on CCAs in 16 emerging market economies from 2001 to 2012. They count the policy changes separately, decomposing them into several categories.

Pandey et al. (2016) study the dynamics of the capital controls aimed specifically at foreign borrowing by Indian firms. They construct a de-jure dataset for this particular class of assets i.e. restrictions on external commercial borrowing by Indian firms. In a more recent study Pandey et al. (2019b) quantify the legal regulations applicable to foreign portfolio investors interested in investing in the Indian financial markets, over the period from 2000 to 2018. The dataset constructed by them discerns information from legal instruments to identify whether the instrument tightens or eases capital controls on investment by foreign institutions in different asset classes such as debt, equity and derivatives.

Some of the older de-jure measures fail to capture the intricacies of capital controls for specific countries owing to their low frequency. This problem becomes more acute for a country like India, where it is possible that within a span of a year, many rules and regulations are imposed by the authorities in order to change the capital account openness. In such a situation some of these de-jure measures, especially those constructed for multiple countries, do not convey an appropriate picture because the value of the indices typically do not change a lot over time. For instance the Chinn-Ito index does not adequately capture the gradual relaxation of controls implemented in India since the mid 1990s because it continues to assign the same score unless all restrictions are removed. This would erroneously imply for example that India's capital account is mostly closed.

Some of the more recent de-jure indices pertain to specific asset classes (such as external commercial borrowing etc) and do not convey a comprehensive picture of capital account changes. In addition, it may be the case that even when de-jure measures show that the capital account has been opened up, in reality foreign capital flows may not change much because the country in question may not be an attractive investment destination perhaps due to weaknesses in its macroeconomic fundamentals or because of other impediments to foreign investment such as cumbersome tax laws etc. These will not get captured by de-jure measures of capital openness. Likewise, while there maybe many legal restrictions in place, investors may find loopholes in them and be able to bring in more capital than what the de-jure measures would imply. In these circumstances, de-facto measures of capital account openness help to understand how open a country's capital account truly is.

There are two categories of de-facto measures of capital account openness: quantity based and price-based. Quantity based measures use actual data on the gross or net flows of capital.

In our paper, we focus on the price-based measures instead, in particular, on the Covered Interest Parity (CIP) which is a pure arbitrage condition. Interest rate parity forms the bedrock assumption of international finance in some sense, and CIP remains the benchmark for detecting departures from perfect capital mobility. As described by Frankel (1992), the condition that Covered interest differential should be zero in the event of perfect international capital mobility is an unalloyed criterion for measuring the degree of financial market integration across national boundaries.

In the absence of transaction costs and barriers to capital movement, capital flows equalize interest rates across countries when contracted in a common currency. However, in countries where the capital account is subject to restrictions or there exist impediments to foreign investment, these two interest rates diverge. This is referred to as the CIP deviation and it captures all barriers to financial integration. In other words, deviations from the CIP provide their own barometer of conditions in the foreign exchange market and can be used to measure how far removed a country is from perfect capital mobility.

CIP deviations can change over time depending upon, among other factors, the gradual changes brought about in capital control actions. The advantage of such a measure is that it captures all complexities in capital account related restrictions as reflected in the prices and is also a dynamic measure which evolves in keeping with the degree of financial integration.

For a country such as India whose capital account is never fully open or entirely closed but instead the restrictions on foreign capital entry or exit undergo periodic and frequent changes, CIP deviations might be able to accurately reflect the changes over time in the extent of India's financial integration.

In our paper, we analyse the CIP deviations between the offshore non-deliverable forward (NDF) currency market and the onshore spot currency market to deduce the trend of capital account openness in India over a long period of time.¹ We explore variations in the CIP deviation and connect the same with documented changes in underlying capital control actions.

Our study is similar to Hutchison et al. (2012a) who examine the effectiveness of international capital controls in India by analysing the daily return differentials between the onshore Rupee market and the offshore NDF market. They argue that the de-jure and de-facto capital controls in India have varied over the 1998-2011 period but a general trend of liberalization is clearly evident. While their study ended in 2011 we extend the time period to 2019 to study the more recent developments.

In the aftermath of the Global Financial Crisis of 2008, many changes were implemented in India with regard to the CCAs. As shown in Pandey et al. (2019b), if we just look at foreign portfolio investment, since 2000 there have been at least one CCA every year. This hints at the elaborate framework of CCAs that is in place in India. They also find that a greater extent of easing of capital controls took place in the post 2008 period particularly post 2013. This necessitates the need to analyse capital account openness in India in the more recent years. While their measure is a de-jure one, to the best of our knowledge no study has documented whether increased relaxations of legal restrictions has indeed translated into greater capital account liberalisation in India in the post 2008 period.

Moreover, while Hutchison et al. (2012a) divide their sample period into exogeneously motivated sub-periods based on their constructed dataset of CCAs, we let the data uncover the gradual evolution of capital account openness through multiple sub-periods. We then connect each of these sub-periods with the corresponding changes in CCAs as documented in the existing

¹There exist both onshore and offshore markets for trading in Rupee. The dominant segment of the offshore market is the non-deliverable forward (NDF) market.

studies to build a comprehensive and coherent narrative about capital account liberalisation in India over the past two decades.

Among other studies, Ma and McCauley (2013) compare the evolution of India and China's capital account openness based on 8 de-facto measures for the period 2000-2012. They found India to be more financially open than China on 6 out of the 8 measures. One of the measures constructed and analysed in their study is the gap between the onshore interest rate and the implied interest rate inferred from the offshore NDF market, similar to Hutchison et al. (2012a).

The rest of the paper is structured as follows. In section 2, we provide a brief background of India's capital account liberalisation process over the past several years. In section 3 we discuss the development of the offshore NDF market for the Indian Rupee. We describe the calculation of the CIP deviation using the offshore NDF implied yield in section 4 and present structural breaks in the CIP deviation in section 5. In section 6 we estimate the no-arbitrage bands and present the results of the SETAR model and finally end with concluding remarks in section 7.

2 India's capital account liberalisation

India adopted liberalisation reforms in the early 1990s prior to which it was primarily a closed economy especially with regard to its capital account transactions. This was part of a broader agenda of reforms initiated after the balance of payments crisis of 1991. On the external front, the reforms included dismantling of trade restrictions, move towards current account convertibility, a market oriented exchange rate regime and a gradual opening up of the capital account. While liberalising the capital account the approach adopted was a calibrated one.

With the Latin American debt crisis of the early 1980s and the Asian financial crisis of 1997 fresh in mind, India prioritised certain kinds of flows and agents in the liberalisation process. In particular, India favoured non debt flows such as foreign direct investment (FDI) and portfolio investment flows over debt flows (Sengupta and Gupta (2015), Mohan and Kapur (2009)). Currently, barring a few sectors, FDI is universally allowed. Portfolio equity flows have also witnessed significant liberalisation. In contrast, debt flows are subject to numerous restrictions including borrowers and lenders having to satisfy eligibility conditions, minimum maturity period, ceilings on interest rate spread and end-use restrictions.

While authorities have been gradually relaxing the legal restrictions that govern foreign investment flows, from time to time new restrictions are also imposed on the foreign investors. As a result, despite the adoption of liberalisation reforms in 1990s, even today in India there exists a comprehensive and complex legal and regulatory framework of capital controls and an extensive array of restrictions on capital account transactions.² As described in Pandey et al. (2016), the capital controls system in India contains innumerable rules tailored to the asset class, investor type, recipient type, transaction magnitude etc. In other words, the capital account liberalisation that has taken place in India over the last couple of decades can be viewed as a continuous process rather than a one off event.

Two recent studies present detailed datasets on de-jure capital control actions (CCAs) taken by the Indian authorities since early 2000s. These datasets have been hand-constructed by studying the legal instruments issued by the authorities and they provide a comprehensive description of the changes that have taken place in CCAs in India over the last couple of decades. The fine-grained dataset constructed by Pandey et al. (2016) throw light on the CCAs pertaining to external commercial borrowing (ECB) by Indian firms. They classify each CCA into 'easing' vs. 'tightening'. Their database has 76 unambiguous CCAs about firms' foreign borrowing between

²See for example, Pandey et al. (2019a), Sengupta and Gupta (2015), Hutchison et al. (2012b), and Shah and Patnaik (2008) among others, for a brief description of India's elaborate capital controls regime.

January 2004 and September 2013. Of these, they find that 68 are easing actions whereas 8 are tightening.

As per their database, the highest number of CCAs occurred in 2012 and 2013, towards the end of their sample period. Majority of these were aimed at easing restrictions on Indian firms. Most tightenings took place in 2007 in response to a surge in capital inflows. This shows in general that in recent years, the Indian authorities have been easing the capital controls and permitting more and more firms to borrow from abroad.

Pandey et al. (2019b) construct a database that quantifies the legal regulations applicable to foreign portfolio investors (FPI) interested in investing in the Indian financial markets. Over a period of 18 years from January, 2000 to December, 2018, their dataset records a total of 151 CCAs. This implies that on average, foreign portfolio investment in India has faced roughly 8 or 9 CCAs annually during the period. Their dataset covers various asset classes such as debt, equity, derivatives etc.

In their dataset, 2018 saw the maximum number of CCAs and the years 2000, 2005 and 2009 saw the least number of such events. They also classify the CCAs into easing and tightening events. FPI easing events denote events that have the effect of relaxation of existing controls or any action that makes it easier for foreign investors to invest in the host country. Conversely, FPI tightening events denote events that have the effect of increasing the capital controls or any actions that make it harder for foreign investors to invest in the host country.

Like the previous study on ECB, Pandey et al. (2019b) find that for the full period of the dataset, the easing events are substantially higher in number at 99, compared to the tightening events which were 27 in number. For all the years, except 2003 and 2006, the number of easing events is higher than the number of tightening events. The maximum number of FPI easing events took place in 2018 (14 in number) followed by 2008 (11), 2013 (10) and 2012 (9). The maximum number of FPI tightening events also took place in 2018 (9) followed by 2008 (4).

These two datasets combined together imply that in recent years there has been a relaxation of capital controls in India on foreign investment flows. It will be interesting to see whether these de-jure actions get reflected in the CIP deviations that we estimate in this paper.

3 The Rupee NDF market

Due to the periodic and frequent imposition of capital account restrictions by the Indian authorities, international participants in cross-border transactions are not able to get easy access to the onshore currency market to either hedge their exposures to the Indian Rupee (INR) or to speculate on the currency movements.³ As a result, over the years an offshore non-deliverable forward (NDF) market has developed. By virtue of being located in financial centres outside India participants in this market can escape the stringent capital account restrictions of India and yet take a speculative position on the expected changes in say the Rupee-Dollar exchange rate. They are also able to hedge their exposures to the INR by accessing these non-deliverable forward contracts.

As described in Guru (2009), an NDF is similar to a regular forward foreign exchange contract. The main difference is that an NDF does not involve a physical settlement of the contract and exchange of currencies. The underlying premise is that the NDF contracts are traded on currencies that are not deliverable offshore. These currencies are from countries such as India that have partial or zero capital account convertibility.

³According to the BIS triennial survey (2019), over-the-counter trades in the Indian rupee accounts for 1.7 percent of the total global forex turnover, compared with 1.1 percent in the 2016 survey.

The Rupee NDF market is largely a USD settled market on the INR-USD rate. The NDF contracts in Rupee are traded in over the counter (OTC) market at various offshore locations such as Singapore, HongKong, London, Dubai and New York. Trading volumes are concentrated in the markets with highest trading time overlap. Singapore is the largest market trading USD-INR offshore.

Alongside the size of India’s open economy, the Rupee NDF market has also grown in size over the years. According to the Bank of International Settlements triennial survey (BIS, 2019), the average daily volume for INR outright forwards stood at USD 62.72 billion in April 2019 compared with USD 29.91 billion in spot market trades. Within this category, NDFs accounted for a significant share of the increase in trading between 2016 and 2019.

The NDF volumes for the USD/INR currency pair reported a staggering three-fold increase, from around USD 16.4 billion in 2016 to USD 50 billion in 2019. The offshore NDF market now accounts for roughly 82 percent of the total outright forwards in USD/INR in 2019 as against 74.3 percent in 2016. India accounts for about 18.2% of the global trade in NDFs (Thorat (2016)). The offshore Rupee market is not only large it is also mostly unregulated.

4 Onshore-Offshore yield

We follow Hutchison et al. (2012a) to calculate the NDF implied domestic interest rate, as follows:

$$r = \frac{F_N}{S}(1 + i_{USD}) - 1$$

where S is the spot exchange rate of the US dollar in terms of the Indian Rupee, F_N is the NDF rate of a certain maturity, i_{USD} is the interest rate on dollar deposits of the corresponding maturity (LIBOR rate). The deviation of the domestic interest rate from the implied domestic yield measures the CIP deviation. We use the Mumbai Interbank Offer Rate, MIBOR as the domestic onshore rate. We focus on one month maturity NDF rate, but also test our results using three months maturity as a robustness check.

We access daily data on NDF contracts and the spot rate from Thomson Reuters Eikon database, and obtain data on 1-month and 3-months LIBOR rates on dollar deposits from the Federal Reserve Board. MIBOR rate on Rupee deposits of same maturity is obtained from the Reserve Bank of India. Our data spans from August 1999 to July 2019. Since both the interest rates are annualized, we first de-annualize the two interest rates, and then compute the resulting r , and re-annualize the interest rate by multiplying by 12. We then compare the implied domestic yield with the MIBOR rate to capture the extent of CIP deviations.

Figure 1 presents the annualized CIP deviations for our sample period. The CIP deviations between 1999 and 2010 ranged between $\pm 10\%$, with outliers around the time of the 2008 Global Financial Crisis. It is worth noting that the size of the CIP deviations reduced

Table 1 presents summary statistics of the CIP deviation, along with the percentage of negative observations. The average CIP deviation for the full sample period is close to -0.73%, while the median is close to half a percentage point.

As can be observed from Figure 1, there are substantial variations in the magnitude of the CIP deviations during the sample period. While one can exogenously identify the periods of substantial variations based on capital control changes, the way it has been done in Hutchison et al. (2012a), this approach introduces subjectivity in the determination of those periods. We

Figure 1 Daily CIP deviations from 1999 to 2019

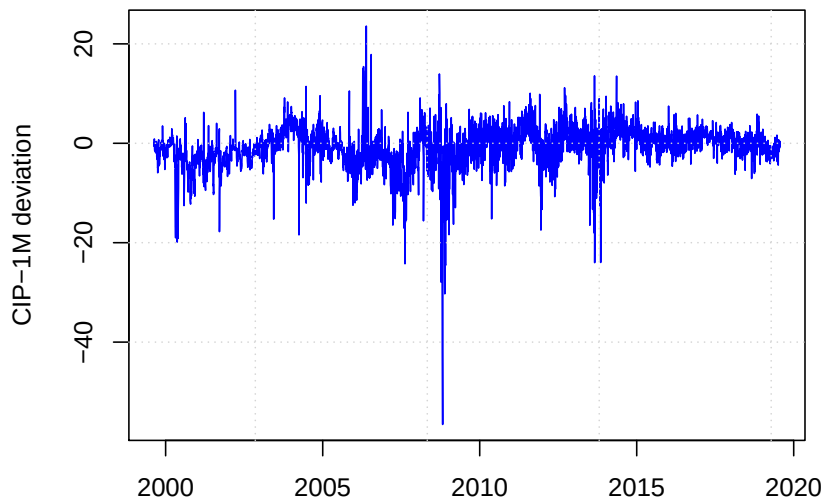


Table 1 Summary statistics of CIP deviations

	CIP deviation (%)
Minimum	-56.56
Q1	-2.49
Median	-0.40
Mean	-0.73
Q3	1.57
Maximum	23.58
% of negative obs.	55.14
# of obs.	4,703

instead let the data divide the sample period into sub-periods based on a structural break test, and then trace the policy changes / capital control measures for the identified sub-periods.

5 Structural breaks in CIP deviation

The methodology for determining structural breaks endogenously is well-established in the literature.⁴ Given a standard linear regression model, the test checks for parameter instability in the model in sub-periods. The approach relies on estimating the breakpoints obtained by minimizing the residual sum of squares in each sub-period. The null hypothesis is that there is no structural break, against the alternative that there are unknown number of breaks. We set the maximum number of structural changes to be five.

The null of no structural break is rejected in favor of the alternative. Figure 2 superimpose the breakpoints on the CIP deviations graph. The five break-dates divide the CIP deviations into six sub-periods. While in the first sub-period (1999-2002) the CIP deviations are mostly negative, in the second sub-period (2002-2005), a large number of CIP deviations are positive. The third sub-period (2005-2009) overlaps with the Global Financial Crisis period, during which we see a large number of negative deviations along with several outliers.

The fourth (2009-2012) and the fifth (2012-2015) sub-periods have a relatively similar range in which the CIP deviations fluctuate with larger number of positive deviations, while the sixth sub-period (2015-2019) has relatively smaller deviations than the previous periods. A negative

⁴See for example, (Andrews, 1993; Bai, 1994; Bai and Perron, 1998, 2003; Zeileis et al., 2003, 2010).

(positive) CIP deviation implies that the domestic onshore rate was lower (higher) than the NDF implied yield.

We also find that the average CIP deviation in the last sub-period, 2015 August to 2019 July is the lowest in our sample, at 0.27%. This is also evident from figure 2 which shows that over time especially from 2015 onwards, the CIP deviation has become smaller, and more tightly distributed around zero.

The narrowing of the CIP deviation implies less barriers to cross-border capital movements and hence, greater capital account openness. Likewise the widening of the deviations in 2005-2009 or 2012-2015 imply that the connectedness of markets and transmission of shocks that we associate with international capital mobility were likely slowed presumably because of imposition of capital controls.

In general we find that the CIP by and large does not hold in India and there have been periods of wide deviations from the interest rate parity in the past 20 years.

Figure 2 Daily CIP deviations from 1999 to 2019

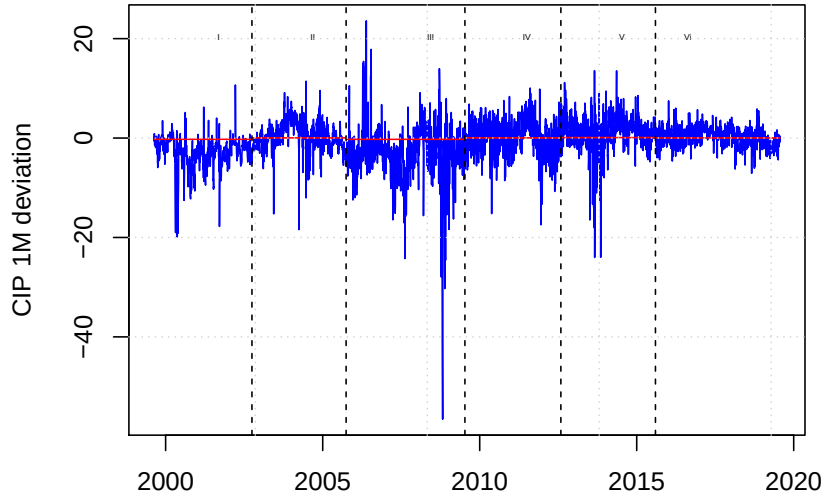


Table 2 provides the summary statistics in each sub-period along with the corresponding dates.

Table 2 Summary statistics of sub-periods CIP deviations

Sub-period	Start date	End date	# of obs.	Median	Mean	SD	% -ve obs.
I	1999-08-18	2002-09-27	1136	-2.43	-2.99	3.19	90.30
II	2002-10-01	2005-09-29	1094	0.27	0.33	2.92	45.67
III	2005-09-30	2009-07-10	1379	-2.77	-3.10	5.96	77.96
IV	2009-07-13	2012-08-01	1115	0.51	0.37	3.47	42.82
V	2012-08-02	2015-08-06	1099	1.46	1.09	3.84	29.28
VI	2015-08-07	2019-07-25	1448	0.38	0.27	1.81	42.09

We find the maximum widening of the CIP deviations in the period from 2005 to 2009, when the deviation in absolute magnitude reached 3.10%. A look at the de-jure data presented in Pandey et al. (2016) shows that the highest number of capital account restrictions on Indian firms' foreign borrowing was in the year 2007. Likewise Pandey et al. (2019b) find that the second highest number of capital controls imposed on foreign portfolio investment by the Indian authorities was in the year 2008. Hence, this seems to be the period when India's capital account

was relatively closed.

The narrowing of the CIP deviation since 2015 also fits well with the recent de-jure data on capital control actions as described in Pandey et al. (2019b). As mentioned earlier, they find a large number of capital control relaxations announced by the Indian authorities in 2017 and 2018. In fact 2018 witnessed the maximum number of easing events since 2000. This hints at the possibility that capital control relaxations in recent years have indeed resulted in greater financial integration of the Indian economy. In ongoing work we plan to further validate this result by examining the data on the actual flows of foreign capital entering the Indian financial markets.

6 No-arbitrage bands estimation

For each sub-period, we estimate the no-arbitrage bands using a three regime threshold auto-regression (TAR) model. The three regime TAR model is based on the number of thresholds inferred from the Hansen’s test which tests for the null hypothesis of no threshold, one threshold and two threshold. The test is conducted sequentially, wherein the null of linearity is tested against the alternative of one threshold. If the null is rejected, a second test is conducted for the null of one threshold against the alternative of two thresholds. Our test results reject the null of in favor of two thresholds in all except the IV and VI regimes. In the IV regime, the null of linearity is rejected in favor of one threshold, while in the VI regime, the null of linearity is not rejected. Based on the test results, we estimate the three regime TAR model for regimes I-V.

The three regime TAR model is given as:

$$\begin{aligned} z_t &= \phi_0^{(1)} + \sum_{k=1}^p \phi_i^{(1)} z_{t-k} + \epsilon_t^{(1)} & z_{t-d} \leq C_{1,t} \\ z_t &= \phi_0^{(2)} + \sum_{k=1}^p \phi_i^{(2)} z_{t-k} + \epsilon_t^{(2)} & C_{1,t} \leq z_{t-d} \leq C_{2,t} \\ z_t &= \phi_0^{(3)} + \sum_{k=1}^p \phi_i^{(3)} z_{t-k} + \epsilon_t^{(3)} & z_{t-d} > C_{2,t} \end{aligned}$$

where, z_t is the time series of interest, in this case, CIP deviations, k indicates the lag structure to be imposed on z_t , $C_{1,t}$ is the lower threshold while $C_{2,t}$ is the upper threshold when z_t is positive. Arbitrage opportunities emerge only when z_t exceeds either $C_{1,t}$ or $C_{2,t}$. ‘d’ is the delay parameter. It determines the lag with which arbitrageurs enter the market. ϵ_t^j is i.i.d $(0, \sigma^{(j)2})$, j indicates the upper ($z_{t-d} \leq C_{1,t}$), middle ($C_{1,t} \leq z_{t-d} \leq C_{2,t}$) or the lower regime ($z_{t-d} > C_{2,t}$).

If arbitrageurs enter the market, the next observations of z_t will move rapidly towards zero. However, when the absolute value of z_t is small, arbitrage will not be profitable and z_t may not change much (Martens et al. (1998)). Thus, in the upper and the lower regime, z_t is expected to follow a stationary process, while in the middle regime, it may follow a non-stationary process.⁵

We estimate the model for each sub-period separately. We use the Akaike Information Criterion (AIC) to determine the lag length. We also need to select the set of delay parameters, S , to be used in the model estimation. We use a variety of values for delay parameters, $S = 1, 2, 3, 5$. Our results are similar across all delay parameters. Hence, we only report the results for $S = 1$.

Table 3 reports the three-regime TAR estimates for sub-periods I, II, III, and V, and a two-regime TAR model for the sub-period IV. The subperiods are identified in Section 5. The

⁵For z_t to be stationary, the sum of the coefficients (ϕ_i^j) in the upper and the lower regime should strictly be less than 1.

table shows the beginning and end dates of each sub period, the selected model (two-regime or three-regime), and the TAR estimates. In the three-regime model, the negative threshold serves as the lower boundary, while the positive threshold serves as the upper boundary of the no-arbitrage bands. We also report the percentage of observations in each regime, the sum of AR coefficients in each regime, and the AR(1) coefficient. Figure 3 plots the CIP deviations along with the estimated thresholds for the lower and high regime.

Figure 3 TAR model estimation results on CIP deviations with estimated boundaries

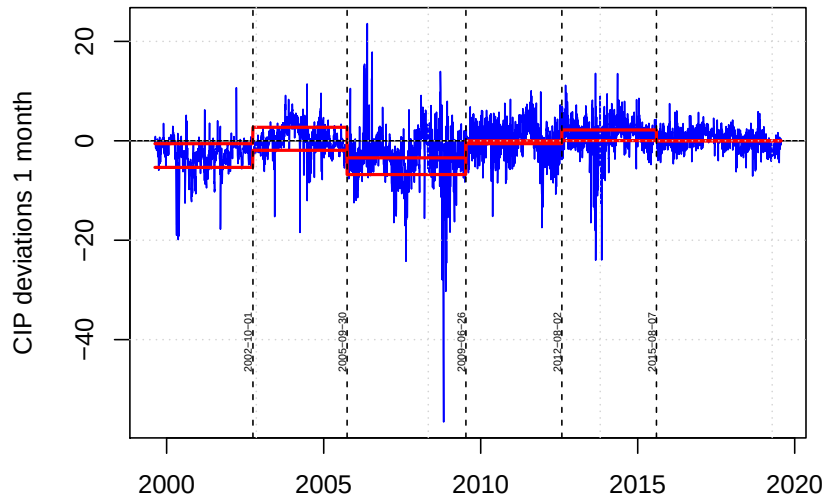


Table 3 indicates that the size of the no-arbitrage bands varies substantially over the five sub-periods. What is worth noting is that the size of the bands declines gradually over time. From a bandwidth of 4.77 in sub-period I, we observe that the bandwidth declines to 2.15 in sub-period V. This decline signals an easing of capital account restrictions and is consistent with our earlier finding that in recent years the CIP deviations have become smaller.

We also observe that in the third sub-period, both the thresholds are in the negative zone. The third sub-period corresponds to the Global Financial Crisis period, when the capital controls were strengthened. The threshold estimated for the fourth regime based on one-regime TAR model is close to zero.

It is also noteworthy, that the sum of the AR coefficients for observations in the middle regime are close to unity, indicating random walk movements within the bands. This is in line with the theory behind the no-arbitrage bands, where the deviations in CIP could not be taken advantage of due to capital controls and transactions costs. This reflects the elaborate and comprehensive system of capital controls that remains in place in India implying that cross-border movement of foreign capital is always subject to conditions and restrictions.

7 Conclusion

Interest parity and arbitrage are age-old concepts in financial economics. In a day and age when a plethora of both de-jure and de-facto measures of a country's capital account openness have been used in the literature without there being any consensus as to which measure is better, a relatively under-utilised measure is a price-based one that uses deviations from the Covered Interest Parity.

In our paper we study the changes in the capital account openness over time of the Indian economy using the CIP deviations. India offers a great case study to analyse CIP deviations,

because it has an elaborate system of capital controls in place and the authorities keep altering the rules on a regular basis thereby changing the underlying conditions for arbitrageurs. Our analysis reveals that India has become more financially integrated with the global markets in recent years, as demonstrated by smaller CIP deviations and narrowing arbitrage bands.

Our finding has important policy implications for India in context of the trade-offs presented by the Impossible Trilemma. With a narrower band, the linkage among currency and money markets is necessarily tighter. Changes in one price or rate are more likely to be transmitted quickly and more predictably into other prices guided by CIP. As an inflation targeting country since 2016, India needs greater monetary policy autonomy. Greater capital account openness over time implies that there is limited room for stabilising currency fluctuations without triggering macroeconomic distortions in the domestic economy.

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Sub-period	Start date	End date	Lower threshold	High threshold	% of obs.			AR(1) coefficient			Sum of AR coefficients			Lag length	AIC	MAPE	Obs.
					Low	Medium	High	Low	Medium	High	Low	Medium	High				
I	1999-08-18	2002-09-27	-5.35	-0.58	0.15	0.68	0.17	0.71	0.64	-0.19	0.54	0.92	0.52	5	1081.61	1.42	732
II	2002-10-01	2005-09-29	-1.91	2.72	0.17	0.62	0.21	0.07	0.41	-0.01	0.31	0.82	0.28	5	1087.40	2.03	705
III	2005-09-30	2009-07-10	-6.79	-3.43	0.17	0.26	0.57	0.78	-0.01	0.26	0.89	1.00	0.55	10	2698.75	3.93	894
IV	2009-07-13	2012-08-01	-0.58		0.34	-	0.66	0.32	-	0.34	0.71	-	0.88	9	1600	3.86	717
V	2012-08-02	2015-08-06	0.04	2.19	0.30	0.32	0.38	0.48	1.12	0.14	0.73	1.42	0.76	9	1624.34	1.69	707

Table 3: TAR model on CIP deviations: Estimation results